# Bipeng Wang

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## SUMMARY

Machine Learning Engineer and AI researcher with a Ph.D. in Chemical Engineering and an M.S. in Computer Science from USC. Experienced in developing deep learning models, multimodal AI systems, LLM-powered applications, and large-scale scientific machine learning pipelines using PyTorch. Published multiple first-author papers and built end-to-end AI solutions spanning research, deployment, and scientific discovery.

## EDUCATION

09/2019-08/2025 **Ph.D. (Chemical Engineering)**, *Univ. of Southern California, Los Angeles, USA*
01/2021-12/2023 **M.S. (Computer Science)**, *Univ. of Southern California, Los Angeles, USA*
09/2017-05/2019 **M.S. (Chemical Engineering)**, *Univ. of Southern California, Los Angeles, USA*
09/2013-05/2017 **B.S. (Applied Chemistry)**, *Shenyang Univ. of Chemical Technology, Shenyang, China*

## PROJECT EXPERIENCE 2019-2026

### AI Agent Systems & LLM Applications, 09/2025-03/2026

- Developed an AI-powered scientific assistant integrating Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), vector databases, and spectroscopy workflows for scientific discovery.
- Built a FastAPI-based backend and interactive web application supporting spectrum upload, automated peak detection, visualization, semantic retrieval, and AI-assisted analysis.
- Implemented retrieval pipelines using Pinecone and sentence-transformer embeddings, enabling semantic search across scientific literature and experimental spectroscopy datasets.
- Integrated LLM reasoning with domain-specific scientific workflows to generate automated summaries, answer user questions, and support scientific decision-making.
- Designed end-to-end AI workflows combining data processing, vector search, reasoning, and report generation, improving usability and reducing manual analysis effort.
- Developed and deployed a full-stack AI system spanning backend services, retrieval infrastructure, machine learning pipelines, and user-facing interfaces.

### Uncertainty Aware Multimodal Emotion Recognition, 01/2024-05/2024

- Developed an uncertainty-aware multimodal transformer combining HuBERT audio representations and VideoMAE visual features for audiovisual behavior understanding.
- Designed scalable PyTorch training pipelines and evaluation workflows, improving F1 score by ~6% over baseline models.
- Implemented uncertainty estimation and confidence calibration mechanisms to improve prediction robustness and interpretability.
- Optimized multimodal feature fusion strategies, improving cross-modal alignment and enhancing performance under noisy real-world conditions.

### Machine Learning Accelerated Scientific Simulation, 09/2019-present

- Developed neural-network surrogate models to accelerate large-scale scientific simulations by over 100× while maintaining high predictive accuracy (<5 meV prediction error).
- Designed symmetry-aware deep learning architectures and scalable PyTorch pipelines for learning complex spatiotemporal representations from atomistic trajectories.

- Implemented ANN, inverse FFT reconstruction, and Bi-LSTM models for long-horizon sequence prediction and sparse temporal signal recovery.
- Applied anomaly detection and uncertainty quantification techniques to identify rare events and improve simulation reliability.
- Published 7 first-author papers on machine learning methods for scientific simulation and time-series modeling.

## SKILLS

**Machine Learning:** PyTorch, Transformers, Deep Learning, Representation Learning, Multimodal Learning

**Generative AI:** LLMs, RAG, Vector Databases, Prompt Engineering, Agentic AI

**Deployment:** FastAPI, REST APIs, Pinecone, Git, Linux

**Programming:** Python, SQL, JavaScript

**Scientific Computing:** NumPy, SciPy, Pandas, Scikit-learn

## PUBLICATIONS

### Machine Learning Model Development:

- **Wang, B. P.** et al. Interpolating Nonadiabatic Molecular Dynamics Hamiltonian with **Artificial Neural Networks**. *J. Phys. Chem. Lett.* **2021**, *12*, 6070-6077. (Link)
- **Wang, B. P.** et al. Interpolating Nonadiabatic Molecular Dynamics Hamiltonian with **Bidirectional Long Short-Term Memory Networks** . *J. Phys. Chem. Lett.* 2023, 14, 7092-7099. (Link)
- **Wang, B. P.** et al. Identifying Rare Events in Quantum Molecular Dynamics of Nanomaterials with Outlier Detection Indices. J. Phys. Chem. Lett. 2024, 15,10384-10391. (Link)

### Scalable Computational Frameworks:

- **Wang, B. P.** et al. Interpolating Nonadiabatic Molecular Dynamics Hamiltonian with **Inverse Fast Fourier Transform**. *J. Phys. Chem. Lett.* 2022, 13, 331-338.(Link)
- **Wang, B. P.** et al. Efficient Modeling of Quantum Dynamics of Charge Carriers in Materials Using Short Nonequilibrium Molecular Dynamics. *J. Phys. Chem. Lett.* 2023, 14, 8289-8295. (Link)
- **Wang, B. P.** et al. Electron-Volt Fluctuation of Defect Levels in Metal Halide Perovskites on a 100 Ps Time Scale. *J. Phys. Chem. Lett.* 2022, 13, 5946-5952. (Link)
- **Wang, B. P.** et al. Sub-Bandgap Charge Harvesting and Energy up-Conversion in Metal Halide Perovskites: Ab Initio Quantum Dynamics. Npj Comput. Mater. 2025 (Link)